

Project Documentation

Auto Ticket Classification using Flow Designer

ServiceNow Automation Project

1. Introduction

- Project Title: Auto Ticket Classification using Flow Designer
- Team Members: Abinaya S - Project Lead & ServiceNow Developer
Afrin U – Business Analyst & Configuration Developer
Keerthika Devi G S - Testing & Documentation Analyst
- Platform: ServiceNow
- Automation Technology: ServiceNow Flow Designer

2. Project Overview

Purpose: The project automates the classification of school IT helpdesk tickets. Newly created tickets are analyzed using keywords in the Short Description/Description, and the appropriate Category and Subcategory are assigned automatically.

Features:

- Automatic classification when a new ticket is created.
- Category and Subcategory assignment based on predefined keywords.
- Dependent-choice logic between Category and Subcategory.
- Email notification to the caller after ticket processing.
- Structured storage of ticket information in the Incident WorkFlow table.
- No-code/low-code maintenance using Flow Designer.

3. Architecture

Application Layer: ServiceNow provides the application interface for creating and managing IT helpdesk tickets.

Automation Layer: Flow Designer triggers when a new record is created in the Incident WorkFlow table. The flow checks whether Category is empty and then evaluates keyword conditions.

Data Layer: The custom Incident WorkFlow table stores Number, Caller, Category, Subcategory, Short Description, Description, State, Assigned Group, and Assigned To.

Classification Mapping:

Keyword / Issue	Category	Subcategory
Wi-Fi / WiFi	Network	Wi-Fi
Projector	Hardware	Projector

Password / Login	Access	Forgot Password
Slow / Hanging	Performance	Slow Computer

4. Setup Instructions

Prerequisites:

- Access to the authorized ServiceNow instance.
- Appropriate permissions to access the Incident WorkFlow table.
- Access to Flow Designer for viewing or configuring the automation.
- A valid caller record/email for notification testing.

Configuration:

1. Create/configure the Incident WorkFlow custom table.
2. Configure the required ticket fields and choices.
3. Configure Category/Subcategory dependent-choice relationships.
4. Create the Flow Designer flow with a Record Created trigger.
5. Configure keyword-based classification branches.
6. Configure the email notification action.
7. Validate the flow using test tickets.

5. Authentication

ServiceNow platform authentication: Access to the application and configuration is controlled through authorized ServiceNow user accounts and platform permissions. Specific authentication credentials are not documented in the project source.

6. User Interface

The user interface is provided through ServiceNow for entering and viewing Incident WorkFlow tickets. Relevant screenshots can include the ticket form, Category/Subcategory fields, and Flow Designer configuration.

7. Testing

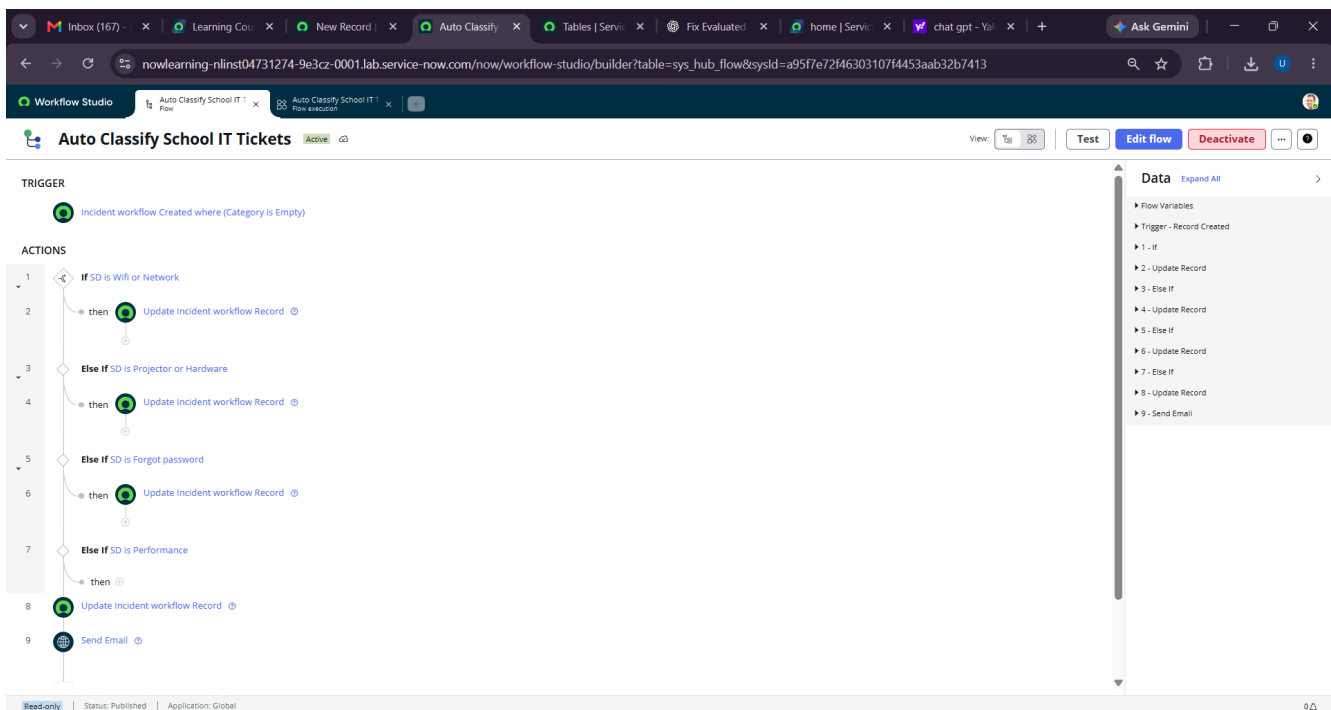
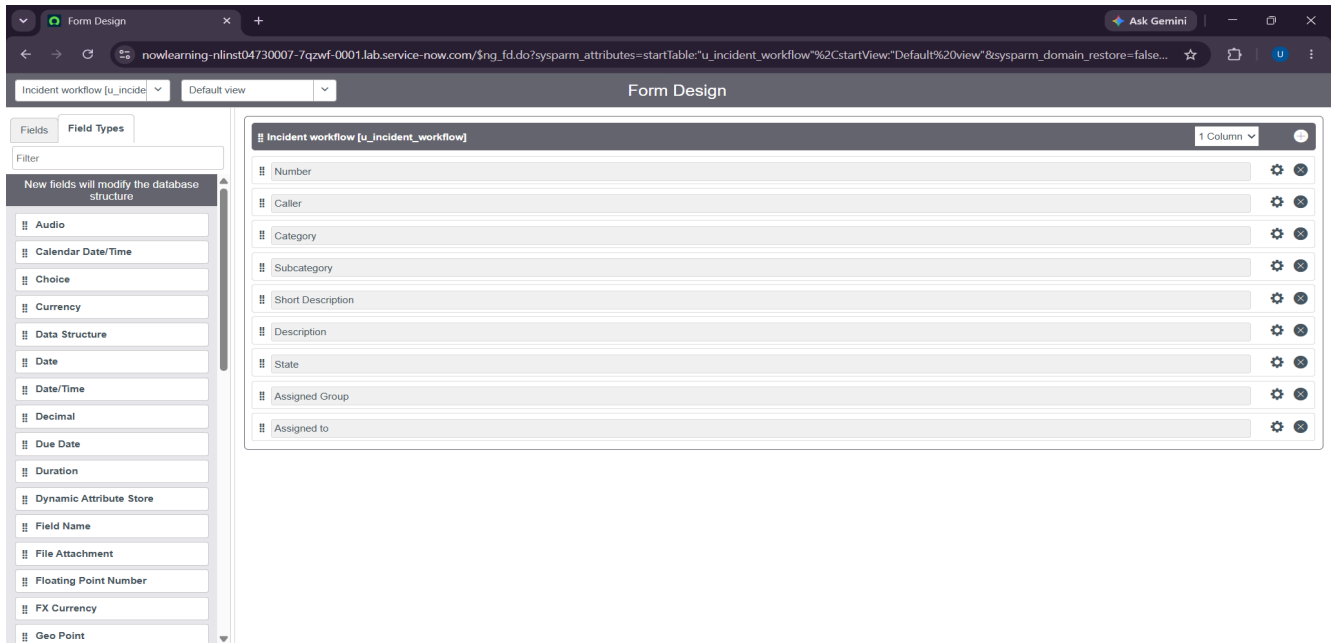
Testing strategy: Functional and performance testing was performed using representative helpdesk tickets.

- Ticket creation test.
- Wi-Fi classification test: Network → Wi-Fi.
- Projector classification test: Hardware → Projector.
- Password/Login classification test: Access → Forgot Password.
- Slow/Hanging classification test: Performance → Slow Computer.
- Email notification test.
- Multiple-ticket processing test.
- Data validation test.

- Unknown/unmatched keyword scenario.

Model testing note: Training and validation accuracy are not applicable because the project uses predefined keyword rules rather than a trained machine-learning model.

8. Screenshots or Demo



Columns Controls Application Access						
Table Columns for text Search						
Dictionary Entries						
	Column label	Type	Reference	Max length	Default value	Display
X	Category	String (Full UTF-8)	(empty)		100	false
X	Subcategory	String (Full UTF-8)	(empty)		100	false
	Updated	Date/Time	(empty)		40	false
X	Number	String	(empty)		40	javascript:global.getNextObjNumberPadded();
X	Subcategory	String	(empty)		40	false
X	Short description	String (Full UTF-8)	(empty)		1,000	false
	Assigned Group	Reference	.NET Application		32	false
X	Category	String	(empty)		40	false
	Created by	String	(empty)		40	false
	Assigned to	Reference	.NET Application		32	false
X	Short Description	String	(empty)		40	false
X	Description	String	(empty)		40	false
X	State	String	(empty)		40	false
X	Subcategory	String	(empty)		40	false
X	Caller	Reference	Abi		32	false
	Sys ID	Sys ID (GUID)	(empty)		32	false
	Created	Date/Time	(empty)		40	false
	Updates	Integer	(empty)		40	false
	Updated by	String	(empty)		40	false
+ Insert a new row...						
Delete Update Delete All Records						

Workflow Studio						
EXECUTION DETAILS Auto Classify School IT Tickets						
Show Action Details						
FLOW STATISTICS						State
Run as: System Administrator						Completed
2026-10-01 06:29:48						132ms
TRIGGER						
Incident workflow Created						
ACTIONS						
1	If SD is Wifi or Network	Flow Logic	Evaluated - True	2026-10-01 06:29:48	1ms	
	Update Record	Core Action	Completed	2026-10-01 06:29:48		
3	Else If SD is Projector or Hardware	Flow Logic	Not Run	2026-10-01 06:29:48	1ms	
	Update Record	Core Action	Not Run	2026-10-01 06:29:48		
5	Else If SD is Forgot password	Flow Logic	Not Run	2026-10-01 06:29:48	0ms	
	Update Record	Core Action	Not Run	2026-10-01 06:29:48		
7	Else If SD is Performance	Flow Logic	Not Run	2026-10-01 06:29:48	0ms	
	Update Record	Core Action	Completed	2026-10-01 06:29:48	6ms	
	Send Email		Completed	2026-10-01 06:29:48	124ms	
ERROR HANDLER						

9. Known Issues

Current documented status: No major defects were identified in the documented UAT scenarios. The classification depends on the configured keyword conditions; tickets containing unsupported or unrecognized wording may require manual review or additional flow branches.

10. Future Enhancements

- Expand the keyword library to support additional helpdesk issue descriptions and variations.
- Add more categories and subcategories as school IT support requirements grow.
- Improve handling of tickets containing multiple issue keywords.
- Add more automated routing and assignment rules.
- Add additional notification and escalation conditions.
- Introduce reporting or dashboards if future project requirements require ticket analytics.
- Add broader validation rules and test scenarios for new classification branches.